

**School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi-110067**

**MCA 2-Years Course Structure and Syllabus)
(From Admissions in Academic Year 2020-21 Onwards)**

Course Structure

The MCA programme would be a four semester programme with minimum 72 credits.

MCA (Semester-I)			
Course	Credits	Type	
Programming in C CS-401	2	Core	Bridge Course for students with B.Sc./ B.Com./ B.A. degree (Non-Computer Science/Non-IT) with Mathematics at 10+2 Level or at Graduation Level.
Computer Fundamentals CS-402	2	Core	
Data Structures CS-403	4	Core	
Discrete Mathematics CS-404	4	Core	
Computer Architecture CS-405	4	Core	
Numerical Methods CS-406	4	Core	

MCA (Semester-II)		
Course	Credits	Type
Database Management Systems CS-407	4	Core
Computer Networks CS-408	4	Core
Design and Analysis of Algorithms CS-409	4	Core
Object Oriented Programming CS-410	4	Core
Artificial Intelligence CS-411	4	Core
Operating Systems CS-412	4	Core

MCA (Semester-III)		
Course	Credits	Type
Software Engineering CS-413	4	Core
Theoretical Computer Science CS-414	4	Core
Elective-I	4	Elective
Elective-II	4	Elective
Elective-III	4	Elective
Elective-IV	4	Elective

MCA (Semester-IV)		
Course	Credits	Type
Project CS-460	8	Core

Elective-I, Elective-II, Elective-III and Elective-IV would be chosen from the list of electives courses given below:

List of Elective Courses	
Microprocessors CS-434	Computer Graphics and Multimedia CS-428
Probability and Statistics CS-440	Machine Learning CS-433
Advanced Data Structure and Algorithms CS-422	Advanced Computer Architecture CS-421
Web Based Programming CS-445	Cloud Computing and IoT CS-426
Compiler Design CS-427	Wireless Networks CS-446
Parallel and Distributed Systems CS-439	Front End Design Technique CS-432
Simulation and Modeling CS-441	Data Mining and Knowledge Discovery CS-430
Mobile Communication and Computing CS-436	Big Data and Analytics CS-424
Mobile Application and Programming CS-435	Cyber Security CS-429
Blockchain Technology CS-425	Natural Language Processing CS-437
Soft Computing CS-442	Digital Image Processing and Computer Vision CS-431
Software Testing CS-443	Agile Methodology and DevOps CS-423
VLSI Design CS-444	Object Oriented Analysis and Design CS-438

SYLLABUS

SEMESTER I

1. Programming in C

An introduction to the process of program design and analysis using the C programming language. Topics to be covered include basic data types and their operators, I/O, control structures (selection, loops), arrays, string, structs and unions, dynamic memory allocation (malloc and free), pointers and pointer arithmetic, and files. Some simple sorting and searching algorithms.

Suggested Readings:

- B.W. Kernighan, Dennis M. Ritchie, The C Programming Language, PHI/Pearson
- Dromey, How to Solve it By Computer, PHI
- E Balaguruswamy, C Programming, Tata Mc Graw Hill
- Stephen G. Kochan, Programming in C, Pearson Education

2. Computer Fundamentals

Logic gates, Boolean algebra, K-map, Map simplification, **Combinational circuit Design** – Half adder, full adders, Decoders, Encoder, Magnitude Comparator, Multiplexer, Read-Only memory, PLAs; **Sequential Circuit Design** – Flip-Flops, Registers, shift registers, Counters, **Processor Logic Design** – Arithmetic circuit, logic circuit, Accumulator, Status register; **Control Logic Design** - Hard-wired control, micro-program sequencer; Digital Integrated circuits – introduction to Bipolar transistor Characteristics, RTL, DTL, I²L, TTL, ECL, MOS, CMOS.

Suggested Readings:

- Moris Mano M, Digital Logic and Computer Design, Fundamentals, PHI 2004
- Givone D D, Digital Principles and Design, Tata McGraw- Hill, 2002

3. Data Structures

Classification of Data Structures: Primitive and Non-Primitive, Linear and Nonlinear; Linear lists, Stack, Linked structures; Orthogonal lists; Multilinked Structures; Tree : Binary tree, Tree traversals, Binary Search trees; Tree deletion; Threaded binary tree; AVL trees; General ordered trees :Expression trees; Huffman trees; B-trees; B+ trees; Forest; Graphs: DFS, BFS, Shortest path algorithm; Introduction,

Fundamentals of the Analysis of Algorithm Efficiency Notion of Algorithm, Fundamentals of Algorithmic Problem Solving, Asymptotic Notations and Basic efficiency classes, Mathematical analysis of Recursive and Non-recursive algorithms. Recursion - Factorial, GCD, Fibonacci Sequence, Tower of Hanoi. Sorting and searching algorithms and their analysis; Queue: Definition, Representation, Queue Variants: Circular Queue, Priority Queue, Double Ended Queue; Applications of Queues. Programming Examples.

Suggested Readings:

- Horowitz Ellis and Sahni Sartaj, “Fundamentals of Data Structures”, W H Freeman and Co.
- Tremblay Jean-Paul and Sorenson Paul G., “An Introduction to Data Structures With Applications”, McGraw-Hill, Inc.
- Kruse Robert L., “Data Structures and Program Design”, Prentice Hall.
- Gilbert Richard F. and Forouzan Behrouz A., “Data Structures: A Pseudocode Approach with C”, Cengage Learning.
- Cormen Thomas H., “Introduction to Algorithms”, The MIT Press.

4. Discrete Mathematics

Sets, Relations, Functions, Logic operators, truth tables, normal forms, Propositional and predicate calculus, Boolean Algebra, Euclidean algorithms, Fibonacci numbers, Congruence and equivalence relation, Groups, Subgroups, semi-group, monoids, Error detecting & correcting codes; permutation & combination, pigeon hole principle, Principle of inclusion and exclusion, Ordinary and exponential generating functions, Recurrence relation; trees, graphs, bipartite graphs, planar & dual BFS & DFS algorithms, Floyd's and Dijkstra shortest path algorithms, Kruskal & Prim's algorithms for minimum cost spanning tree; Regular expression, Finite state machine, Grammars.

Suggested Readings:

- Tremblay and Manohar, Discrete Mathematical Structures, Tata McGraw Hill
- Maggard, Thomson, Discrete Mathematics, 1 st Edition
- Semyour Lipschutz, Varsha Pati “Discrete Mathematics”, IInd Edition Schaum's Series TMH
- Kolman, Busby and Ross, “Discrete Mathematical Structures”: Prentice Hall India, Edition 3
- C.L. Liu, Elements of Discrete Structures. 6. Rosen, Discrete Mathematics and application.

5. Computer Architecture

Data representation – number system, two's complement, Arithmetic operations using 2's and 1's complements, floating-point representation and arithmetic operations; Register Transfer and Micro-operations, instruction codes and formats, addressing modes, timing and control signal generation, instruction cycles, memory reference instructions, Input out instructions; Central Processing unit – stack organization, Data transfer and manipulation, program control; Pipeline and vector Processing–Arithmetic and Instruction Pipeline, vector operations, array processors; Input output organization –I/O verses memory bus, Isolated verses memory mapped I/O, Priority interrupts, DMA, I/O processors; Memory Organization– RAM, ROM, Associative memory, Cache memory, memory management hardware.

Suggested Readings:

- Morris Mano, Computer System Architecture Pearson Education, 2012
- David A. Patterson, John L. Hennessy, Computer Organization and Design: The Hardware/Software Interface, Morgan Kaufmann, 2009
- William Stallings, Computer Organization and Architecture: Designing for Performance, Pearson Education, 2007
- Behrooz Parhami Computer Architectures: From Microprocessors to Supercomputers, , Oxford, 2005.

6. Numerical Methods

Data representation – Fixed-point numbers, Floating-point numbers, Finite data representation; Propagation of error, Root finding' Newton's methods, Fixed-point iteration, Ill-behaved root finding problems, Interpolation, divided differences, Spline functions, Approximation of functions – Chebyshev polynomial; Numerical integration and differentiation – Trapezoidal and Simpson rules, Gaussian numerical integration, Solution of systems of linear equations – matrices equation, method of triangular matrices, Gaussian elimination with pivoting, Jacobson and Gauss Seidel; Ordinary Differential equations – Euler method, Runge-Kutta methods, Multi-step method, System of differential equations.

Suggested Readings:

- M.K. Jain, SRK Iyengar and R.K.Jain, Numerical Methods for Scientific and Engineering Computation
- S.C. Chopra and Raymond P. Canale, Numerical Methods for Engineers
- Sastry, Introductory Methods of Numerical Analysis by
- E.W. Cheney and D.R.Kincaid,. Numerical Analysis

SEMESTER II

1. Data Base Management Systems

Introduction; Database Architecture; Database Analysis: Entity Relationship Model, Database Design: Relational Model, Integrity Constraints, Functional, Transitive, Multivalued and Join Dependency; Normalization: 1NF, 2NF, 3NF, BCNF, 4NF, 5NF; Lossless Join Decomposition and Dependency Preservation; Canonical Cover; Query Languages: Relational Algebra, Tuple and Domain Relational Calculus, SQL: DDL, DQL, DML, DCL; Transaction Processing, Concurrency Control and Recovery Techniques; Advances in Database Systems and Technologies.

Suggested Readings:

- Silberschatz, A., Korth, H.F., Sudarshan, S., Database System Concepts, McGraw-Hill International Edition, 2006 (5 th Edition)
- Elmasri, R., Navathe, S.B., Fundamentals of Database Systems, Fourth Edition, Pearson Education,
- Desai, B.C., An Introduction to Database Systems, Galgotia Publications,
- Date, C.J., An Introduction to Database Systems, Pearson Education, 7 th Edition
- Garcia-Molina, H., Ullman, J.D., Widom, J., Database Systems: The Complete Book, Pearson Education, 2002

2. Computer Networks

Overview of Computer Network; Data Communication – Analog and digital communications, signal and data, Channel characteristics, Nyquist theorem, Shannon's formula, modulations, encoding schemes; Error Detection – VRC, LRC, CRC, Checksum; Transmission media; Multi-channel communication – multiplexing techniques, Multiple access techniques; Switching techniques – Telephone systems, ATM; Queuing Models – little's theorem, Poisson Process, Markov Chain model, M/M/1, M/M/S/K, M/G/1 queues; ALOHA systems; Local Area Networks – CSMA/CD, Token Ring, Token Bus, FDDI, Routing – spanning tree, Bellman-Ford, Dijkstra algorithms, Distance vector and link state routing; IP protocol-fragmentation, reassembly; Transport Protocol – stop-and-wait, Sliding window protocols, congestion control, TCP, UDP; Application Protocols – FTP, Email.

Suggested Readings:

- Andrew S. Tanenbaum, Computer Networks, Fourth Edition, Pearson Education,
- Behrouz A. Ferouzan, Data Communications and Networking, 2nd edition, TMH,
- Alberto Leon-Garcia and Indra Widjaja, Communication Networks, 2 nd Ed, Tata McGraw-Hill,
- William Stallings, Data and computer Communications, 7 th Edition, Pearson Education,

3. Design and Analysis of Algorithms

Some efficient algorithms; Algorithm analysis: worst case and average case analysis, Recurrences and asymptotics; Algorithm design: divide-and-conquer, branch-and-bound, backtracking, dynamic programming, greedy methods; Efficient data structures, amortized analysis, NP completeness; Additional topics: graph problems, string searching, parallel computing, low level techniques for efficient programming.

Suggested Readings:

- T Cormen, C Leiserson, R Rivest, C Stein, Introduction to Algorithms, PHI.
- V. Aho, J. Hopcraft, J. Ulmann, The Design and Analysis of Computer Algorithms, Addison Wesley.
- E Horowitz, S Sahni, S Rajasekaran, Fundamentals of Computer Algorithms, Universities Press
- S. Basse, A. V. Gelder, Computer Algorithms: Introduction to Design and Analysis, Pearson Education Asia Pvt. Ltd.

4. Object Oriented Programming

The Key Attributes of Object-Oriented Programming, The Java Development Kit, The Java Class Libraries. Java's Primitive Types, Literals, The Scope and Lifetime of Variables, operators, Shorthand Assignments, Type conversion in Assignments, Using Cast. Control Statements, Classes, Objects and Methods, Controlling Access to Class Members, Pass Objects to Methods, Method Overloading, Overloading Constructors, Recursion, Inheritance Basics, Member Access and Inheritance, Constructors and Inheritance, Using super to Call Superclass constructors.

Suggested Readings:

- H.M. Deitel, P.J. Deitel, Java : how to program, Fifth edition, Prentice Hall Publication.
- Herbert Schildt, The Java 2: Complete Reference, Fourth edition, TMH.

5. Artificial Intelligence

Overview of AI; Problem solving; various search strategies, Game playing: MinMax, Alpha-Beta pruning; Introduction to logic programming, inference mechanisms; Knowledge representation: Logic, rules, semantics, frames, Conceptual dependency; Uncertain knowledge and reasoning; Hill Climbing, Simulated Annealing, Best First Search: Graphs, Heuristic Functions, A* Algorithm, AND –OR graphs, AO* Algorithm. Expert systems.

Suggested Readings:

- Rich and Knight, Artificial Intelligence, Tata McGraw-Hill.
- Russell and Norvig, Artificial Intelligence: A Modern Approach (3rd Edition), Prentice Hall.
- Nilsson, Artificial Intelligence: A New Synthesis, Morgan Kaufmann Publishers, Inc.
- Winston, Artificial Intelligence, Addison Wesley.

6. Operating Systems

Overview, Computer System Structures, Operating System Structures, Process and Process Scheduling, CPU scheduling, Process Synchronization, Semaphores, Monitors, Deadlock and methods of deadlock handling, Address Space, Paging, Segmentation, Virtual Memory, Page Replacement Algorithms, File and Directory Concepts, Access Methods, File System Allocation methods, Secondary Storage Scheduling and Management, Protection, Security

Suggested Readings:

- Silberschatz, P. Galvin and G. Gagne, Operating System Concepts, 9th Edition
- William Stallings, Operating Systems: Internals and Design Principles, 7th Edition

SEMESTER III

1. Software Engineering

Introduction, Software Product and Process, Software Process Models, Requirements Engineering, Requirements Analysis –Data Flow Diagram, Requirement Specification, Requirement Validation; Design- Concepts, Coupling, Cohesion, Mapping Analysis to Design, User Interface Design, Structure Charts, Coding Principles, Coding Standards and Guidelines, Software Testing Techniques and Strategies, Software Debugging, Software Project Metrics and Estimation Techniques – Empirical, Heuristic and Analytical Techniques, Software Quality Assurance, CASE Tools, Software Maintenance and Reengineering.

Suggested Readings:

- Pressman, R., Software Engineering – A Practitioner's approach, Sixth Edition, McGrawHill International Edition.
- Sommerville, I., Software Engineering, Sixth Edition, Pearson Education.
- Ghezzi, C., Jazayeri, M., Mandrioli, D., Fundamentals of Software Engineering, Second Edition, Pearson Education.
- Jalote, P., An Integrated Approach to Software Engineering, Second Edition, Narosa Publishing House.

2. Theoretical Computer Science

Mathematical reasoning; logic; methods of proof; sets, relations, function; induction principle, Grammars, Chomskian hierarchy of languages and their properties; Finite state machines, Kleen's theorem, Moore and Mealy models; Pushdown automata; Turing machine; Recursive and recursively enumerable sets models, recursive functions; Halting problem, Church's thesis.

Suggested Readings:

- Linz, Peter, An introduction to Formal Languages and Automata, Narosa Publishing House, 2007
- Lewis, H.R., and Papadimitriou, C.H., Elements of Theory of Computation, Pearson Education, 2002
- Hofcroft, J.E., and Ullman, J.D., Introduction to Automata Theory, Languages and Computation, Narosa Publishing house 2008
- Krithivasan, Kamala, Introduction to Automata Theory, Languages and Computation, Pearson Education, 2009
- Martin, J.C., Introduction to Languages and Theory of Computation, Tata McGraw-Hill Publication, 3e, 2007

Elective Courses

1. Microprocessors

Basic Microprocessor architecture, bus structure and memory maps, memory systems for microprocessor, hardware and software interrupts, I/O systems Introduction to Intel family of microprocessor, 8085/8086 microprocessor - architecture, Instruction set, programming.

Suggested Readings:

- Ramesh Gaonkar, Microprocessor Architecture, Programming and Applications with the 8085, Wiley Eastern Limited, 2002
- Barry B. Brey, C.R.Sharma, The Intel Microprocessors: Architecture, Programming and Interfacing, Pearson Education, 2005
- Sunil Mathur, Microprocessor 8086: Architecture, Programming and Interfacing, PHI, 2011
- Douglas V. Hall, Microprocessor and Interfacing, McGraw Hill Company, 2005.